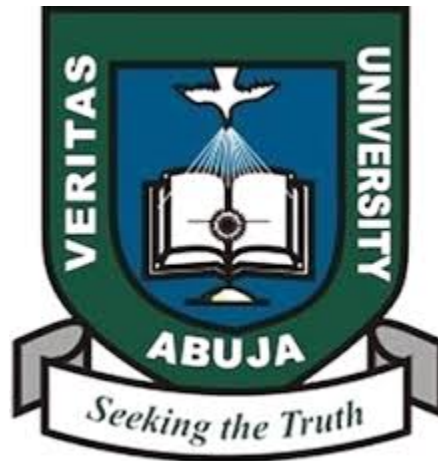


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‘LAB MANUAL’
Course Code: VUA-SEN182
Course Title: “Web Programming (HTML, CSS, Git”)

Prepared by: Mr. Victor Ayamba

Course Contents

Contents	CLOs	Lecture	Practical
HTML: Basics, Elements, Attributes, Headings, Paragraphs, Formatting, Links, Head, Images, Tables, Lists, Blocks, Layout, Forms, Multimedia Tags, Semantic HTML, HTML5 Structure.	CLO – 1 & 2	4	8
CSS: Introduction, Syntax, Selectors (ID & Class), Styling Backgrounds, Text, Fonts, Links, Box Model (Border, Outline, Margin, Padding), Grouping/Nesting, Dimension, Display, Positioning, Floating, Align, Navigation Bar, Image Gallery, Flexbox, Grid Layout, Responsive Design.	CLO – 1 & 2	3	6
JavaScript: Introduction, Output, Statements, Comments, Variables, Data Types, Objects, Functions, Operators, Comparisons, Conditions, Loops, Error Handling, DOM Introduction and Manipulation, Number, String, Date, Array, Window, Screen, Navigator, Popup Alerts.	CLO – 1 & 2	4	8
Bootstrap: Introduction to Bootstrap Framework, Grid System, Containers and Layout, Responsive Design Principles, Typography and Utilities, Navigation Bars, Buttons, Cards, Forms and Input Groups, Tables, Alerts, Modals, Responsive Components and Layout Customization.	CLO – 1 & 2	4	8
Total		15	30

INTRODUCTION TO THE COURSE:

The Internet and the World Wide Web have become fundamental components of modern society, influencing communication, education, business, and everyday life. As digital platforms continue to expand, the ability to design and develop functional and visually appealing web applications has become an essential skill for software engineers and technology professionals. Understanding how web technologies work together enables students to create interactive systems that can serve real-world needs.

This laboratory course provides a practical introduction to modern Web Programming using core client-side technologies such as **HTML, CSS, JavaScript, and the Bootstrap framework as well as version control tools like Git and Github**. Students will learn how to structure web pages using HTML, style and layout interfaces with CSS, enhance interactivity through JavaScript, and build responsive, mobile-friendly user interfaces using Bootstrap. Through hands-on laboratory exercises, students will progressively develop the skills required to design, implement, and maintain modern web applications.

The lab sessions are structured to guide students step-by-step through the development of a meaningful web project while introducing essential web development concepts and tools. Each experiment builds upon the previous one, enabling learners to gradually transform basic web pages into fully structured and responsive websites. At the end of each lab session, tasks are provided to reinforce understanding and encourage independent exploration. Students are advised to follow this lab manual sequentially and carefully adhere to the provided guidelines to gain maximum benefit from the course.

Lab 3: HTML Lists - Demonstrated by Instructor

HTML Lists

HTML lists allow web developers to group a set of related items.

Types of HTML Lists:

- Unordered List
- Ordered List
- Description List

Unordered HTML List

An unordered list starts with the `<ul>` tag. Each list item starts with the `<li>` tag. The list items are marked with bullets by default.

Example:

```
<ul>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>
```

Ordered HTML List

An ordered list starts with the `<ol>` tag. Each list item starts with the `<li>` tag. The list items are numbered by default.

Example:

```
<ol>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>
```

HTML Description Lists

A description list is a list of terms with descriptions.

- `<dl>` defines the list
- `<dt>` defines the term
- `<dd>` describes the term

Example:

```
<dl>
  <dt>Coffee</dt>
  <dd>- black hot drink</dd>
  <dt>Milk</dt>
  <dd>- white cold drink</dd>
</dl>
```

HTML List Tags

Tag	Description
	Defines an unordered list
	Defines an ordered list
	Defines a list item
<dl>	Defines a description list
<dt>	Defines a term
<dd>	Describes the term

Unordered List - List Item Markers

The list-style-type property defines the marker style.

Types:

- disc (default)
- circle
- square
- none

Disc Marker (Default)

```
<ul style="list-style-type:disc;">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>
```

Circle Marker

```
<ul style="list-style-type:circle;">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>
```

Square Marker

```
<ul style="list-style-type:square;">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>
```

No Marker

```
<ul style="list-style-type:none;">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>
```

Nested HTML Lists (Unordered)

Lists can be placed inside other lists.

Example:

```
<ul>
  <li>Coffee</li>
  <li>Tea
    <ul>
      <li>Black tea</li>
      <li>Green tea</li>
    </ul>
  </li>
  <li>Milk</li>
</ul>
```

Horizontal List with CSS

Lists can be styled horizontally to create navigation menus.

Example:

```
<ul>
  <li><a href="#">Home</a></li>
  <li><a href="#">News</a></li>
  <li><a href="#">Contact</a></li>
  <li><a href="#">About</a></li>
</ul>
```

Ordered List - Type Attribute

The type attribute defines numbering style.

Types:

- 1 → Numbers (default)
- A → Uppercase letters
- a → Lowercase letters
- I → Uppercase Roman
- i → Lowercase Roman

Numbers

```
<ol type="1">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>
```

Uppercase Letters

```
<ol type="A">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>
```

Lowercase Letters

```
<ol type="a">
  <li>Coffee</li>
```

```
<li>Tea</li>
<li>Milk</li>
</ol>
```

Uppercase Roman Numbers

```
<ol type="I">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>
```

Lowercase Roman Numbers

```
<ol type="i">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>
```

Control List Counting

Use the start attribute to change the starting number.

Example:

```
<ol start="50">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>
```

Nested HTML Lists (Ordered)

Example:

```
<ol>
  <li>Coffee</li>
  <li>Tea
    <ol>
      <li>Black tea</li>
      <li>Green tea</li>
    </ol>
  </li>
  <li>Milk</li>
```


HTML Description Lists (Again)

Example:

<dl>

<dt>Coffee</dt>

<dd>- black hot drink</dd>

<dt>Milk</dt>

<dd>- white cold drink</dd>

</dl>

Lab 3: Introduction to HTML and Basic Web Page Structure – Handling LISTS

SECTION B: Lab Tasks on LISTS

Task 1: Create an Unordered List

What to do

Create an unordered list to display common student items.

Suggestion

Use the and tags.

Example:

```
<ul>
  <li>Notebook</li>
  <li>Pen</li>
  <li>Textbook</li>
</ul>
```

Task 2: Create an Ordered List

What to do

Create an ordered list showing steps for student registration.

Suggestion

Use the tag.

Example:

```
<ol>
  <li>Visit the school portal</li>
  <li>Fill in your details</li>
  <li>Submit the form</li>
</ol>
```

Task 3: Create a Description List

What to do

Create a description list explaining common school terms.

Suggestion

Use `<dl>`, `<dt>`, and `<dd>`.

Example:

```
<dl>
  <dt>Lecture</dt>
  <dd>- A teaching session</dd>
  <dt>Assignment</dt>
  <dd>- A task given to students</dd>
</dl>
```

Task 4: Change List Item Markers

What to do

Create unordered lists using different marker styles:

- Circle
- Square
- None

Suggestion

Use the `list-style-type` property.

Example:

```
<ul style="list-style-type:square;">
  <li>Item 1</li>
  <li>Item 2</li>
</ul>
```

Task 5: Create a Nested List

What to do

Create a list showing categories and subcategories.

Suggestion

Place a list inside another list.

Example:

```
<ul>
  <li>Fruits
    <ul>
      <li>Apple</li>
      <li>Orange</li>
    </ul>
  </li>
  <li>Vegetables</li>
</ul>
```

Task 6: Create a Horizontal Navigation Menu

What to do

Create a simple horizontal menu using a list.

Suggestion

Use CSS to style the list.

Example:

```
<ul>
  <li><a href="#">Home</a></li>
  <li><a href="#">About</a></li>
  <li><a href="#">Contact</a></li>
</ul>
```

Task 7: Use Different Ordered List Types

What to do

Create ordered lists using different numbering styles:

- Numbers
- Uppercase letters
- Lowercase letters
- Roman numerals

Suggestion

Use the **type** attribute.

Example:

```
<ol type="A">
  <li>Item 1</li>
  <li>Item 2</li>
</ol>
```

Task 8: Control List Counting**What to do**

Create an ordered list that starts from a number other than 1.

Suggestion

Use the `start` attribute.

Example:

```
<ol start="10">
  <li>Item 1</li>
  <li>Item 2</li>
</ol>
```

Task 9: Create a Nested Ordered List**What to do**

Create an ordered list with sub-items.

Suggestion

Nest one `<ol>` inside another.

Example:

```
<ol>
  <li>Main Item
    <ol>
      <li>Sub Item</li>
    </ol>
  </li>
</ol>
```

Task 10: Combine Different List Types

What to do

Create a page that uses:

- Unordered list
- Ordered list
- Description list

Suggestion

Combine all list types in one section.

Example:

```
<ul>
  <li>Item A</li>
</ul>

<ol>
  <li>Step 1</li>
</ol>

<dl>
  <dt>HTML</dt>
  <dd>- HyperText Markup Language</dd>
</dl>
```

Task 11: Emphasize Important Items

What to do

Highlight important list items using formatting tags.

Suggestion

Use `<strong>` or `<em>`.

Example:

```
<ul>
  <li><strong>Important:</strong> Submit your assignment</li>
  <li>Attend lectures</li>
</ul>
```

